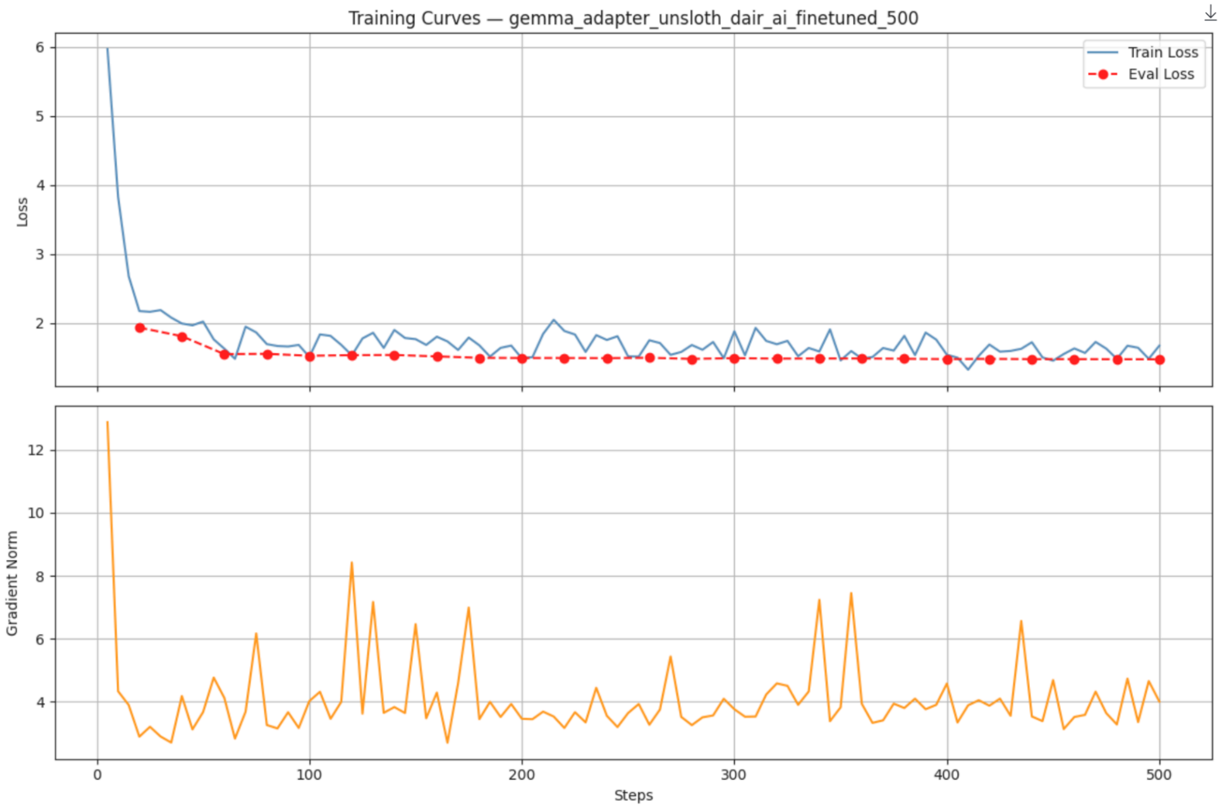
Report

BaseModel vs Finetuned Model vs Finetuned Model (1st layer ablated) vs ………vs Finetuned model(Final layer ablated)

Dataset: dair_ai/emotion dataset → multiclass emotion classification dataset
Model: gemma 3 270M model

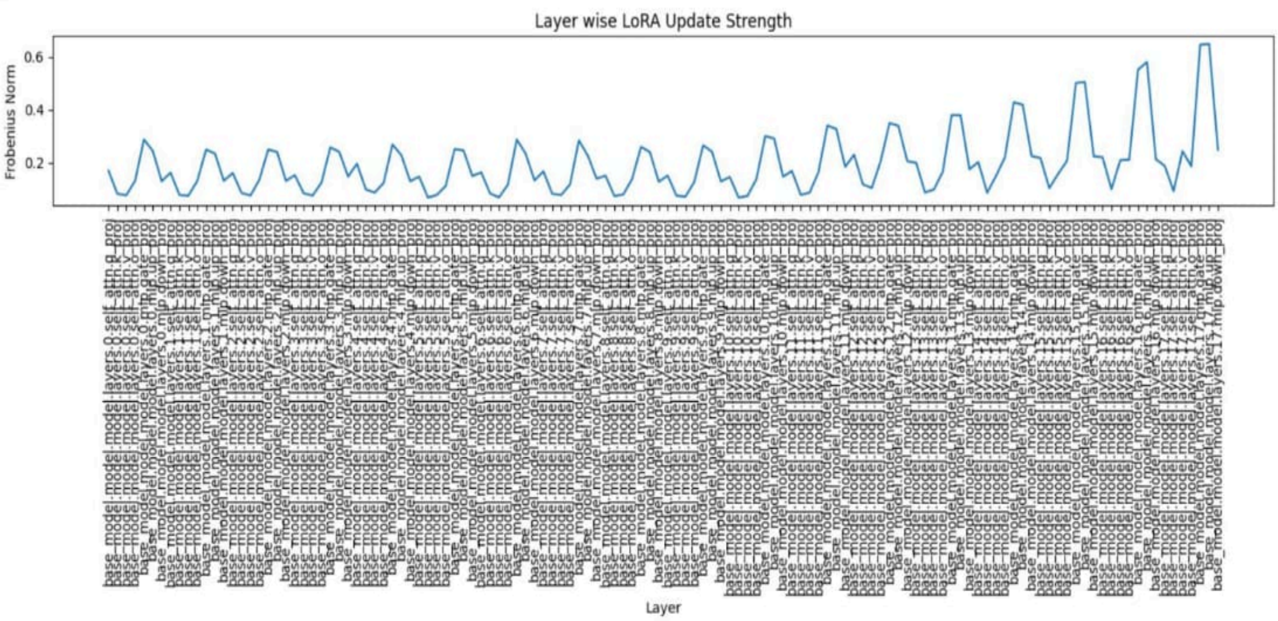
I trained the model for 500 steps . It converged at 200-300 steps .

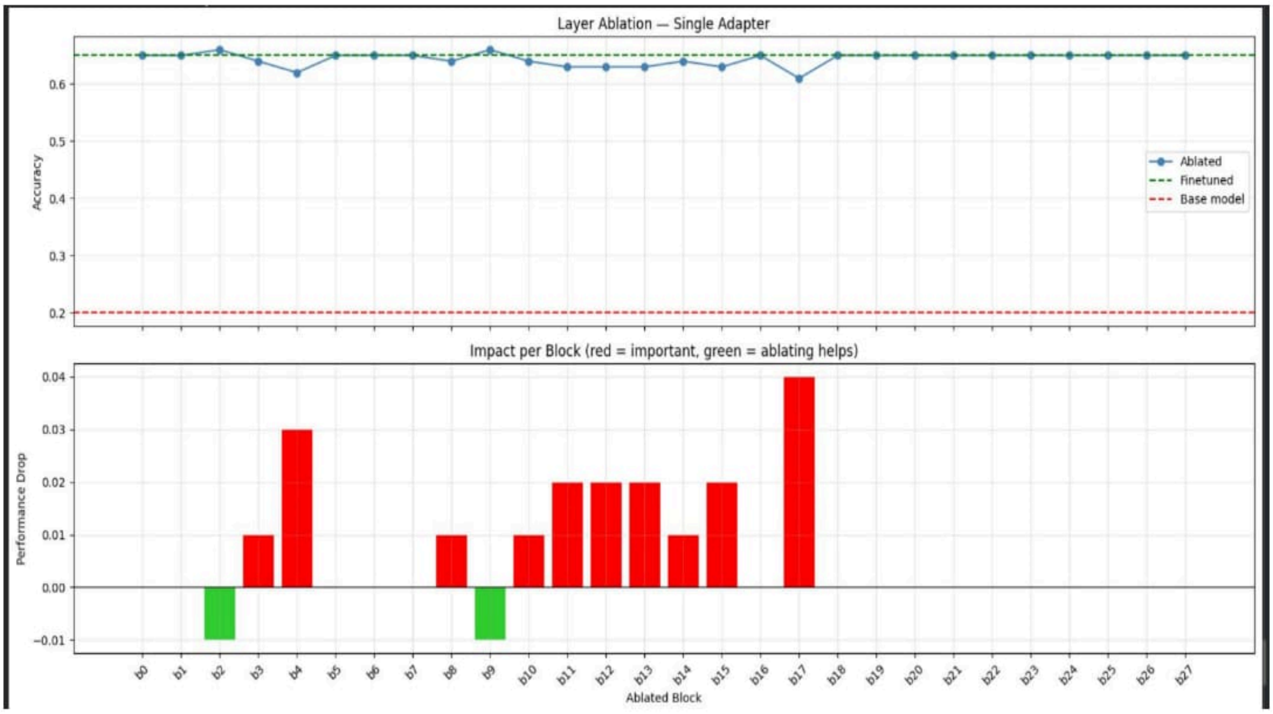


Results:

- **Base Model had accuracy of 0.2**

- **Finetuned model(with all layers) had accuracy 0.65**





Observations:

1) **Norms are low and flat for early blocks and then spikes in blocks 15-17.**

2) **Block 17 has the highest norm spike and ablating it will give a performance drop.**

3) **The blocks 0,1,5,6,7 and 16 have near zero drop so these block updates contribute essentially nothing to emotion classification. But when they all got dropped there was a drop of 0.04**
That means these layers form a redundant but collectively important group. Individually they appear unimportant, but together they carry information needed for emotion classification

4) **Blocks 2 and 9 when ablated separately had a performance increase of 0.01 and when ablated together had performance increase of 0.02**

- **So these layers might have learnt task conflicting updates**
- **Also their contributions are independent.**

5) **When layers 5 and 16 were ablated together, there was a performance increase of 0.01 and when ablated separately there was no performance drop or increase.**

6) **When blocks [3,4,8,10,11,12,13,14,15,17] got ablated there was a drop of 0.29**
This is where the task lives.
- When layer 17 alone was ablated , there was a drop of 0.04

- So layer 17 seems very important layer

- But when layers [3,4,8,10,11,12,13,14,15] (only 17 not included) there was a drop of 0.32
- So even when layer 17 was included in the model, there was a drop of 0.32 , the highest drop.